

RESEARCH ARTICLE

Blind Interleaver Recognition Using Deep Learning Techniques

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ABSTRACT In digital communication systems, channel encoders and interleavers play a crucial role in mitigating the random and burst errors introduced by noisy channels. While information about the encoder and interleaver is typically available in cooperative scenarios, non-cooperative military communication systems often lack such knowledge. This paper explores the application of deep learning to recognize four different interleavers such as block, convolutional, helical, and random especially in non-cooperative environments. By utilizing convolutional neural network and dilation residual network models, we perform the recognition task with input data for interleavers encoded using six different channel encoders such as block, convolutional, Bose-Chaudhuri-Hocquenghem, Reed-Solomon, low density parity check, and polar. Across all cases, both convolutional neural network and dilation residual network models consistently achieve classification accuracy exceeding 95% at varying signal-to-noise ratio values. In addition, we compare the classification accuracy and testing/training time of both our proposed models (i.e. convolutional neural network and dilation residual network) with the recurrent neural network, feedforward network, and autoencoder models available in the literature. Our findings reveal that the accuracy improves with high input sample length and yielding superior results for both the models. Notably, among the five models, dilation residual network outperforms other models in terms of classification accuracy and convolutional neural network outperforms dilation residual network in terms of training/testing time. Our results show that the convolutional neural network and dilation residual network models outperform the other methods. We also compare the classification accuracy of the proposed model with the existing rank-based approach proposed in the literature. Our findings indicate that at the same bit error rate values, our model demonstrates superior classification accuracy.

INDEX TERMS Blind recognition, channel encoders, convolutional neural network (CNN), deep learning, dilation neural network (DRN), interleavers, military communication, non-cooperative scenarios.

I. INTRODUCTION

Recent advancements in digital storage and communication systems have significantly transformed our daily lives. In a digital communication system, the roles of forward error correcting (FEC) codes and interleavers at the transmitter are pivotal, effectively mitigating the random and burst errors, respectively [1]. The information about the FEC

encoder and interleaver employed at the transmitter is crucial for successful decoding at the receiver. In the instances where receivers possess prior knowledge of the encoder and interleaver, decoding and deinterleaving processes are facilitated. However, for scenarios where such prior information is lacking, the need arises for the blind estimation of the channel encoder and interleaver [2]. This blind estimation becomes essential to enable effective decoding in the absence of prior knowledge, underscoring the dynamic nature of modern communication systems. Nevertheless,

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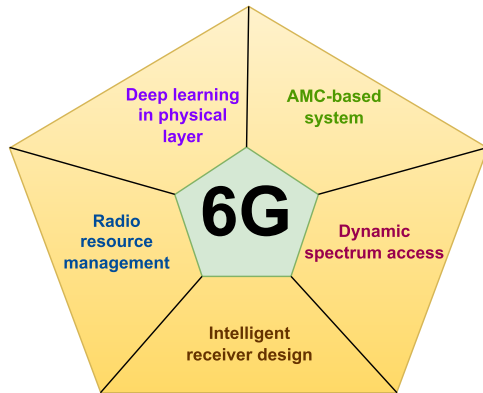


FIGURE 1. Significance of blind recognition techniques in the context of 6G communication.

these blind estimation techniques introduce novel challenges that demand effective solutions and tackling these challenges become critical in diverse applications, particularly in non-cooperative military communication scenarios [3]. During the transition period from 4G to 5G and eventually to 6G, various wireless systems and use cases may employ different cellular standards. Effective cellular operations and radio resource management might necessitate identifying multiple cellular signals (4G, 5G, and 6G) [4] using intelligent or blind receiver systems. Recent progress in deep learning (DL) offers numerous algorithms for recognizing the modulation, channel coding, and interleaving schemes of incoming RF signals across various wireless cellular standards (from 3G to 6G). As 6G communication systems aim to integrate DL techniques into the physical layer, the design and development of intelligent receivers will become crucial. Adaptive modulation and coding (AMC) techniques are very useful in advanced wireless communication systems, as they improve overall spectral efficiency. Moreover, efficient usage of the spectrum is necessary not only to meet current, ever-increasing demands in 5G networks, but also for 6G networks. The modulation and code parameters are usually communicated using control channel to the receiver in AMC-based wireless systems. This approach, however, incurs some capacity loss. Therefore, the blind recognition techniques will result in conserving the channel resources [1], [3], which in turn will further enhance the spectral efficiency of AMC-based wireless systems, playing a vital role in 6G and beyond networks. In 5G/6G, dynamic spectrum access enables cognitive radios to optimize spectrum usage by adapting to real-time network conditions. This improves spectral efficiency, which is crucial for supporting high data rates in next-generation networks. The blind recognition of modulation, channel coding, and interleaving in dynamic spectrum access refers to the ability of a cognitive radio system to identify and adapt to the modulation schemes and channel coding methods being used in the network without prior knowledge. This capability allows the system to dynamically and efficiently utilize the available spectrum

by recognizing the communication parameters in real-time, optimizing performance, and minimizing interference with other users. In summary, the significance of blind recognition techniques in the context of 6G communication is shown in Fig. 1. At the transmitter side, three primary operations are executed before transmitting any digital information: FEC encoding, interleaving, and modulation. In a non-cooperative scenario at the receiver side, understanding all three operations become essential without explicit knowledge. To address this, various traditional approaches in the literature provide blind estimation methods for the modulation scheme [5], [6]. The utilization of interleaving at the output of the FEC encoder is widely employed in digital video and audio broadcasting systems, aiming to enhance performance in the presence of burst errors [7].

Our proposed models have practical applications in non-cooperative military systems, where dynamically changing coding and interleaving parameters during transmission pose challenges for parameter estimation at the receiver end. Representing a significant advancement in signal intelligence (SIGINT) capabilities for military operations, our model excels in intercepting and classifying enemy communications, even amidst diverse interleaving and channel coding techniques. This enhances intelligence gathering significantly. However, implementing our model in real-world scenarios poses few challenges. Rigorous training and testing in real-world conditions, optimization of hardware with limited resources, and mechanisms for continuous learning are essential considerations to maximize its utility and impact while ensuring relevance and performance in dynamic operational environments.

A. MOTIVATIONS

- In numerous practical applications, precise knowledge of coding and interleaver parameters is typically available at the receiving end. Nevertheless, in scenarios characterized by non-cooperation, such as military and spectrum surveillance applications, receivers may lack prior information about coding and interleaver parameters. Consequently, the need arises for blind or semi-blind recognition methods to ascertain these parameters, facilitating in-depth analysis and enhancing adaptability in diverse communication environments.
- As per the summary given in Table 1, the existing research has primarily concentrated on the recognition of FEC codes using deep learning techniques or the identification of specific interleaver parameters through algebraic methods. However, there exists a gap in exploring deep learning for the classification of interleavers. Our motivation is to utilize deep learning for the classification of interleavers and evaluate the accuracy under various SNR conditions.

B. CONTRIBUTIONS

- Our contribution involves the classification of four different interleavers, each distinguished by unique

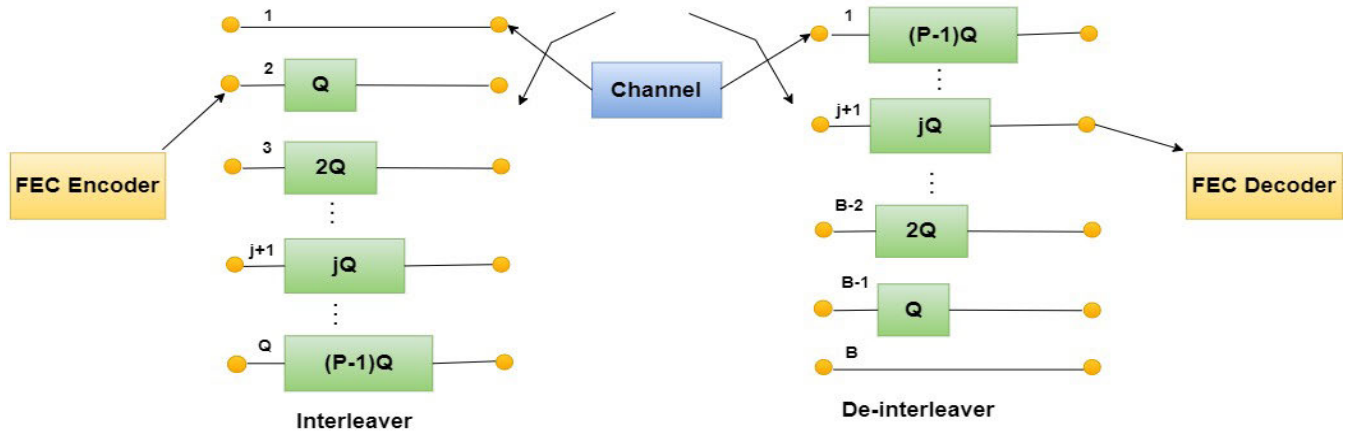


FIGURE 2. Convolutional interleaver/deinterleaver architecture with interleaver depth P and width Q illustrating commutator switches registers to store new symbols and shift out old ones for transmission.

...	m_2	m_1	j_3	j_2	j_1	g_3	g_2	g_1	d_3	d_2	d_1	a_3	a_2	a_1
...	...	k_3	k_2	k_1	h_3	h_2	h_1	e_3	e_2	e_1	b_3	b_2	b_1	0
...	...	...	l_1	i_3	i_2	i_1	f_3	f_2	f_1	c_3	c_2	c_1	0	0

FIGURE 3. Example illustrating convolutional interleaved encoded data sequence $(a_1, a_2, a_3, b_1, b_2, b_3, \dots)$ with parity bits highlighted in red, where the interleaved data will be retrieved column-wise, i.e. $(a_1, 0, 0, a_2, b_1, 0, a_3, b_2, c_1, \dots)$ for transmission with $P = 3$ and $Q = 1$.

parameters. The identification of these interleavers has been carried out using input data coded with the help of different FEC encoders: block, convolutional, Bose-Chaudhuri-Hocquenghem (BCH), Reed-Solomon (RS), low density parity check (LDPC), and polar.

- CNN and DRN architectures of deep learning are used for interleaver classification task at different SNR values assuming additive white Gaussian noise (AWGN) channel.
- The performance of the proposed deep learning models is assessed in terms of their classification accuracy (i.e. probability of correct identification) across various input sizes and time complexity.
- Finally, we have also compared the performance of the proposed CNN and DRN models with various deep learning models such as autoencoder, feedforward neural network (FNN), and recurrent neural network (RNN) as well as with rank-based methods available in the literature.

C. ORGANISATION OF THE MANUSCRIPT

The paper is structured as follows: Section II discusses various works that are available in the literature relevant to the proposed work. Section III introduces the various interleavers and details on the proposed deep learning models

for interleaver recognition, which include neural network architectures, principles, dataset creation, and model training. Section IV presents the simulation results and the relevant observations on the obtained results. Finally, Section V concludes the work, summarizing findings and suggesting future research directions.

II. RELATED WORKS

Various blind estimation techniques for channel encoder and interleaver parameters are available in the literature [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21]. Previous investigations into blind recovery of convolutional encoders have been documented, as seen in references such as [1] and [2], where algebraic and dual-code properties serve as the basis. An alternative method, presented in [9], introduces an algebraic approach to identify puncturing patterns and parent code parameters of punctured convolutional encoders. Further, [10] discussed the blind reconstruction techniques reliant on soft-decision outputs, including average log-likelihood ratio (LLR) of the syndrome posteriori probability (SPP) and average likelihood difference (LD) of parity checks. Notably, these methods assume a predetermined candidate set of convolutional encoders known at both the transmitter and receiver, which is a reasonable assumption, particularly in AMC-based systems.

TABLE 1. Literature Summary.

Reference	Method	Type of interleaver/FEC encoder/modulation recognition details
[1]	Rank deficiency difference estimation	Block coded, convolutional coded, uncoded symbols, and block interleaver
[2]	Dual code	Convolutional encoder
[5]	Graphic constellations and distributed network (DBN)	Modulation schemes: BPSK, QPSK, 8PSK and 16QAM
[6]	high-order cumulants	Modulation schemes: PSK and QAM
[7]	Deep learning	Modulation recognition: PSK, FSK, and FSK
[8]	Decentralized learning and ensemble learning	24 different analog and digital modulators
[9]	Algebraic method	Punctured convolutional encoder
[10]	Average likelihood difference	Convolutional and LDPC encoders
[11]	Iterative expectation-maximization (EM) algorithm	Turbo encoder
[12]	Least square cost function	Systematic convolutional sub-codes of turbo encoder
[13]	Gauss-Jordan elimination through pivoting (GJETP)	Turbo convolutional encoder
[14]	GJETP	Reed-Solomon (RS) encoder
[17]	GJETP	Convolutional and helical interleavers
[19]	GJETP	BCH-coded convolutional interleaver
[20]	Probabilistic framework	Hamming-coded random interleaver
[22]	Deep learning architecture: RNN	BCH, LDPC, turbo, and convolutional encoders
[23]	Deep learning architecture: CNN	Linear block, convolutional, and turbo encoders
[24]	Deep learning architecture: CNN	Convolutional encoders

Additional research in [11] and [12] delved into blind reconstruction of a parallel concatenation of two recursive systematic convolutional (RSC) encoders using iterative expectation-maximization (EM) and least-square methods. Furthermore, in [13], novel algorithms for parameter estimation in turbo convolutional codes were proposed, outlining techniques for reconstructing non-identical RSC encoders and block interleavers in noisy environments. The works of [14] and [15] introduced parameter estimation algorithms specifically tailored for RS codes, capitalizing on the count of non-zero columns and non-zero elements in the column echelon form of the data matrix. Exploring various methodologies for predicting block interleaver parameters, [16] presented diverse approaches. Notably, the technique in [17] was based on the rank deficiency of the data matrix to discern interleaver parameters, while [18] utilized hamming weight distribution for the same purpose. The investigation highlighted in [19] delves into blind identification of convolutional interleaver parameters under noisy conditions. Additionally, in a non-cooperative context, the algorithms for blind estimation of interleaver parameters were developed as detailed in [20].

In recent literature, the application of deep learning techniques for the blind identification of various FEC codes was explored. Recognition of diverse encoders, including Bose-Chaudhuri-Hocquenghem (BCH) codes, low density parity check (LDPC) codes, turbo codes, and convolutional codes, achieved up to 99% accuracy utilizing recurrent neural networks (RNNs) as demonstrated in [22]. Additionally, in [23], a convolutional neural network (CNN) model was proposed and it demonstrated the ability to classify linear block codes, convolutional codes, and turbo codes solely based on noisy information streams. Notably, at low

signal-to-noise ratio (SNR), the CNN model exhibited superior accuracy compared to Gauss-Jordan elimination through pivoting (GJETP), with minimal differences observed at high SNR. Subsequent work in [24] focused on the identification of convolutional codes using the CNN model, achieving accuracy above 90% across 16 different classes of convolutional codes.

III. DEEP LEARNING-BASED BLIND INTERLEAVER RECOGNIZER

In the operation of a block interleaver, the coded symbols are received in a block form from the encoder. The interleaver proceeds to rearrange these symbols by shuffling, after which the reorganized symbols are directed to the data modulator. This shuffling process involves populating the columns of an M -row by N -column ($M \times N$) matrix with the encoded sequence. Once the array is fully populated, the symbols undergo transmission to the modulator in a sequential manner, being fed one row at a time over the channel. Upon reception, the deinterleaver executes the reverse process, with symbols being entered row-wise and extracted column-wise to reconstruct the original sequence.

In the (P, Q) convolutional interleaver setup as stated in [25], the encoded data stream is sequentially placed into a series of Q registers. Each of these registers, starting from the first, has a delay line of dimensions $J \times Q$. As new coded symbols arrive, a commutator switches to the next register in sequence. This operation allows the latest incoming symbol to be stored, while the oldest symbol is simultaneously shifted out for transmission. The deinterleaver, operating synchronously with the interleaver, reverses this process at the receiver end. The architecture of the (P, Q) convolutional

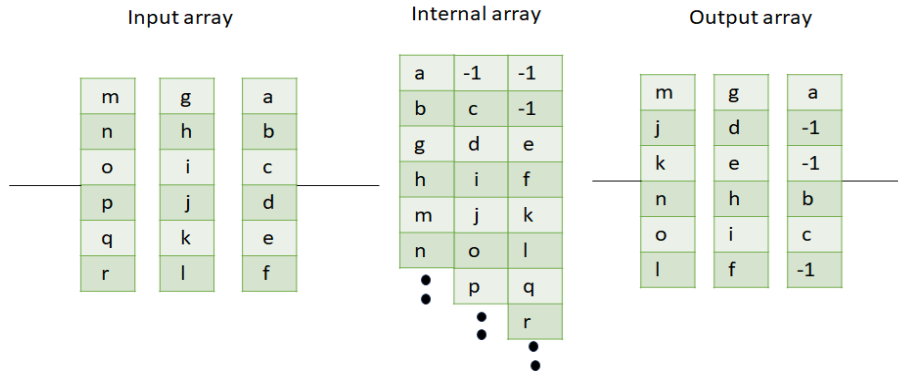


FIGURE 4. Helical interleaver operation illustrating input, internal, and output arrays considering the data sequence as $(a, b, c, d, e, f, \dots, r)$ with $C = 3$, $L = 2$, and $h = 1$.

interleaver and deinterleaver, depicted in Fig. 2, illustrates this mechanism.

A 3×3 convolutional interleaver is utilized for the interleaving process. Consequently, the interleaver depth is denoted as $P = 3$, with a step size of $Q = 1$. It is important to emphasize that the maximum interleaver delay always remains below the codeword length n_c . The interleaver period is determined as $n_c \times P = 3 \times 3 = 9$. Each coded data, represented by letters such as $a, b, c, \dots$, comprises 3 bits, delineated as a_1, a_2, a_3 . Visualizing the convolutional interleaved data within the interleaver can be depicted as illustrated in Fig. 3. The codewords are arranged from right to left, adhering to the alphabetical sequence of “ $a, b, c, d, e, \dots$ ”, wherein the parity check bits are visibly highlighted in red. During transmission, the data is sequentially retrieved column by column, following the right-to-left sequence “ $a_1, 0, 0, a_2, b_1, 0, a_3, b_2, c_1, \dots$ ”.

A helical interleaver is a type of block interleaver that arranges the coded symbols in a spiral or helix pattern within a two-dimensional array. The helical interleaver module reorganizes symbols within an input signal by arranging them in a helical pattern and subsequently transmitting rows of this array to the output port [26]. Internally, the module employs an array with C columns and an infinite number of rows, where C represents the number of columns in the helical array. With a group size parameter denoted as L , the module processes an input of length $C \times L$ during each time step, dividing it into consecutive groups of L symbols. Following the modular reduction operation, the module places the i^{th} group in the array along $\{i \bmod C\}^{\text{th}}$ column. This placement is considered helical due to the modulo operation and the first symbol in the i^{th} group occupies $\{1 + \{(i - 1) \times h\}\}^{\text{th}}$ row, where h signifies the helical array step size parameter. The unoccupied positions in the array retain default contents specified by the initial condition parameter. The module outputs $C \times L$ symbols by sequentially reading the next L rows from the array. At any given time step, the output symbols may stem from the initial condition parameter, the current input vector,

or symbols retained in the array from a prior time step. In Fig. 4, the helical interleaver operation, comprising input, internal, and output arrays, is illustrated with an example assuming $C = 3$, $L = 2$, and $h = 1$. The initial block configuration reveals a matrix with the initial condition parameter initialized to -1 and populated with input symbols as per the helical array structure. Upon receiving the first input vector $[a, b, c, d, e, f]'$, the output at the initial time step is structured as $[a; -1; -1; b; c; -1]$. Subsequently, the array undergoes transformation with the arrival of the second input vector $[g, h, i, j, k, l]'$, resulting in an updated configuration and an output sequence of $[g; d; e; h; i; f]$. Finally, with the third input vector $[m, n, o, p, q, r]'$, the block's internal array evolves further, producing an output of $[m; j; k; n; o; l]$. This process demonstrates the dynamic interplay between input symbols and the specified array configuration, showcasing the interleaver operation over the initial three time steps. In a helical interleaver, the -1 placeholders represent positions where no input symbols are placed. These positions occur because the helical pattern skips certain locations in the array. Therefore, it is expected that few input symbols will not be present in each output block, and instead, they are replaced by -1 in accordance with the helical interleaving pattern. This skipping of positions is inherent to the helical interleaving process and ensures a specific arrangement of the input symbols within the block.

The random interleaver introduces randomness by rearranging data using a state parameter derived from a dynamic process like the clock function. Random interleavers are widely employed in digital communication systems to combat channel fading and burst errors. In satellite communication systems, random interleavers are crucial for mitigating the effects of signal degradation due to atmospheric disturbances [27].

A. INTERLEAVER RECOGNITION PROCESS

The architectural framework of the deep-learning-based blind recognition system for classifying interleavers is depicted in Fig. 5. Starting from the transmitting terminal, a set

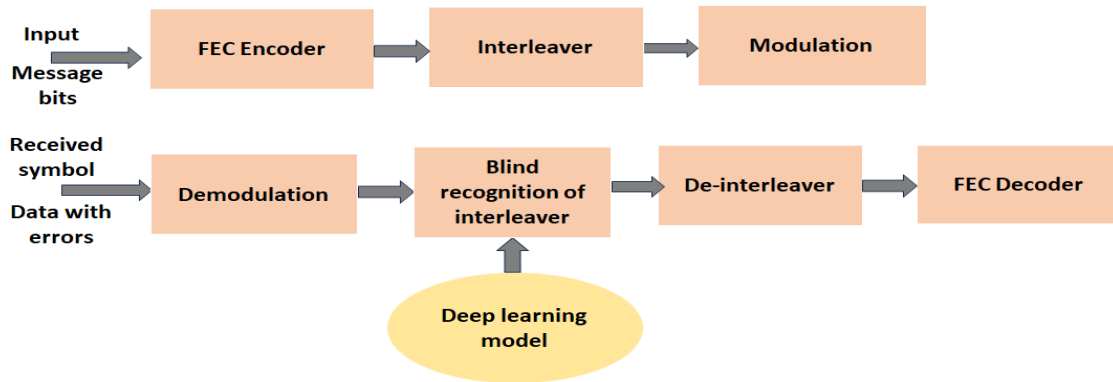
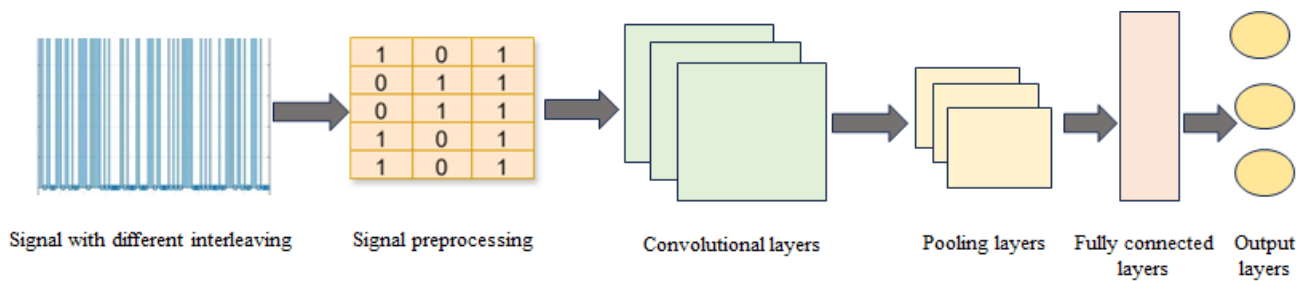
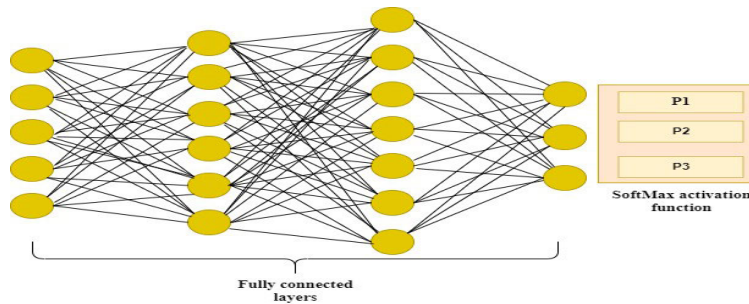


FIGURE 5. Block diagram of interleaver recognition system involving various vital steps to identify interleavers from FEC encoded and interleaved data sequence with errors using deep learning technique.



(a) 1-D CNN model architecture: Unveiling the network structure.



(b) Neural network with FC layers and Softmax for classification.

FIGURE 6. 1-D CNN architecture and key characteristics.

of randomly generated information bits, denoted as $b = [b_1, b_2, \dots, b_k]$, is introduced into the channel encoder. This encoder handles the continuous information bits k and yields an encoded sequence, $c = [c_1, c_2, \dots, c_n]$, where each c_i and b_i belong to the binary Galois Field $GF(2)$. The code rate r is expressed as k/n . The resulting encoded digital sequence initially undergoes modulation and is subsequently transformed into a continuous signal before passing through the communication channel. In the context of this paper, it is assumed that the signals undergo binary phase-shift keying (BPSK) modulation at the transmitter and the received signals undergo corresponding BPSK demodulation and achieve perfect frame synchronization.

It is important to note that similar outcomes can be achieved when employing more advanced modulation

schemes. Upon reaching the receiving terminal, the received data stream is directed to the demodulator for the extraction of demodulated soft data bits, denoted as $y = [y_1, y_2, \dots, y_n]$. Subsequently, the data bits have to be sent to the deinterleaver and FEC decoder for extracting the information bits. This research primarily emphasizes on the autonomous identification or classification of interleaver using CNN and DRN models among four different interleavers, i.e. block interleaver, convolutional interleaver, helical interleaver, and random interleaver, which are having wider applications in digital communication systems such as cellular system, digital video broadcasting (DVB) systems, interleave-division multiple access (IDMA) systems, etc [28]. It is also to be noted that in this paper identification or classification of modulation scheme or FEC encoder has not been discussed, as it is outside

the scope of the proposed work. It is assumed that the receiver knows the modulation scheme and FEC encoder used at the transmitter through modulation classification [3] and FEC encoder classification techniques [22], which are discussed in the introduction section (i.e. literature survey).

B. SUGGESTED NEURAL NETWORK-BASED APPROACHES FOR BLIND IDENTIFICATION

Extensive research in the literature has indicated that neural networks exhibit remarkable performance in classification tasks [22], [23], [24]. The selection of various neural networks parameters, including the number of hidden layers, training algorithm, neuron count per layer, cost function, activation function, and other relevant settings, is conducted judiciously, taking into consideration factors such as algorithm convergence time, minimum validation error, and overall network accuracy, among others. There are several types of neural networks available in the literature like feedforward neural networks (FNN), CNN, RNN, DRN, long short-term memory networks (LSTM), gated recurrent unit (GRU), autoencoders, generative adversarial networks (GANs), etc., each designed for specific tasks and data types [29]. Among them, CNN, DRN, RNN, and autoencoder are best suited for binary classification task. In our work, we have employed the CNN and DRN architectures for our interleaver classification task and compared the accuracy of the proposed architectures with the RNN and autoencoder architectures. In the field of deep learning, CNN stands out as one of the most renowned and frequently utilized algorithms, as evidenced by numerous studies [30], [31], [32], [33], [34], [35], [36], [37], [38]. A commonly used type of CNN, which is similar to the multi-layer perception (MLP), consists of an input layer, an output layer, along with multiple convolutional layers, pooling layers, and FC layers as depicted in Fig. 6a. Pooling layers follow the convolutional layer. The primary objective of pooling is to downsample the feature maps, leading to computational efficiency by reducing the number of trainable parameters. Following the pooling layer, the activation layer introduces non-linearity by applying a non-linear activation function, such as rectified linear unit (ReLU), to the pooled outputs. As shown in Fig. 6b, a softmax function is employed after the fully connected layer for classification tasks to provide a probabilistic interpretation of the model's output, making it suitable for determining the most likely class among the multiple classes. Note that P_1 , P_2 , and P_3 represent the probability values obtained after the softmax function.

The DRN is a type of deep neural network architecture designed for semantic segmentation tasks, particularly in computer vision applications. It utilizes both dilation and residual connections to improve the efficiency and performance of semantic segmentation models. It is designed to excel in image classification, particularly when preserving fine spatial details is critical. It achieves this by incorporating dilated convolutions, which expand the

receptive field of higher layers without compromising spatial resolution. By removing subsampling in selected layers and applying dilated convolutions strategically, DRN strikes a balance between retaining spatial information and capturing contextual cues. This innovative architecture is well-suited for recognizing objects of varying sizes and complexities in images, making it a valuable tool in modern computer vision applications. The foundation of our approach draws inspiration from the network architectures introduced by Lu et al. in their seminal work [39]. The learning process of this network is same as CNN learning process. These architectures are organized into five distinct groups of convolutional layers. In the initial layer of each group, downsampling is achieved through striding, meaning that the convolutional filter selectively operates on even rows and columns. The dilated convolution operation can be mathematically expressed as

$$y(i) = \sum_k x(i + rk) \cdot w(k). \quad (1)$$

Here, x is the input, w is the filter, y is the output, i is the index, and r is the dilation rate.

The residual block, inspired by residual network (ResNet) architecture, includes a shortcut connection to address the vanishing gradient problem. The output of the residual block is given by

$$\text{O/P} = \text{ReLU}(\text{Conv}(W_2, \text{ReLU}(\text{Conv}(W_1, \text{Input}) + S))). \quad (2)$$

Here, W_1 and W_2 are the learnable parameters (weights) of the convolutional layers, $\text{Conv}(\cdot)$ represents the convolutional operation, and S is the shortcut connection. The DRN architecture combines dilated convolutions and residual connections in a hierarchical manner. The overall architecture is a composition of multiple residual blocks, where some of the convolutional layers have dilation rates greater than 1, which can be expressed as

$$\text{DRN} = \text{R-Block}_1 \circ \text{D-Conv}_1 \circ \text{R-Block}_2 \circ \text{D-Conv}_2 \circ \dots \quad (3)$$

Here, $\circ$ represents the composition of functions and R-Block and D-Conv denote the residual block and dilated convolutional layers, respectively.

C. MODEL DESIGN AND TRAINING

A common structure of a CNN includes an input layer, several hidden layers, and an output layer. In particular, these hidden layers frequently incorporate three fundamental layers: convolutional layers, pooling layers, and FC layers. In this paper, we have adapted the architecture of AlexNet [40] to modify the network presented in [41], resulting in the creation of a 1-D CNN, as detailed in Table 2. These modifications encompass three primary aspects. Initially, as our dataset consists of 1-D sequences formed by demodulated encoded streams, we have conducted a 1-D reconstruction of the

AlexNet, replacing the original 2-D convolutional and pooling layers used for image processing with their 1-D counterparts. Subsequently, in order to maintain parameter uniformity, we have reduced the number of feature maps in the last four layers by half and adjusted the neuron counts in the two FC layers to 128 and 64. Lastly, to enable the classification of interleavers, we have modified the last FC layer of the 1-D CNN to contain 12 neurons.

TABLE 2. Architecture specifications of the CNN.

Layer	Step size	Output size
Input	/	16384×1
Convolutional + ReLu	$11 \times 1/4$	4094×96
Maxpooling	$3 \times 1/2$	2046×96
Convolutional + ReLu	$5 \times 1/1$	4094×128
Maxpooling	$3 \times 1/2$	1022×128
Convolutional + ReLu	$5 \times 1/1$	1022×192
Convolutional + ReLu	$5 \times 1/1$	1022×192
Convolutional + ReLu	$5 \times 1/1$	1022×128
Maxpooling	$3 \times 1/2$	510×128
Global Average Pooling	/	128
Dense + ReLu	/	128
Dropout(0.5)	/	128
Dense + ReLu	/	64
Dropout(0.5)	/	64
Dense + Softmax	/	12

TABLE 3. Architecture of DRN at dilation rate [2, 4, 6, 8].

Layer	Configuration
Input type	Shape: (4096,1) and (8192,1)
Convolutional (Initial)	Filters:64, Kernel Size: 11, Strides: 4
Maxpooling	Pool size: 3, Strides:2
Residual Block ($\times 4$)	Filters: 256, Kernal Size: 3, Activation: Relu,
Maxpooling	1022×128
Convolutional + ReLu	1022×192
Convolutional + ReLu	1022×192
Convolutional + ReLu	1022×128
Maxpooling	510×128
Global Average Pooling	128
Dense + ReLu	128
Dropout(0.5)	128
Dense + ReLu	64
Dropout(0.5)	64
Dense + Softmax	12

TABLE 4. Interleaver types and sizes.

Interleaver Type	Size
Block (M, N)	(3,2), (4,2), (3,3), (4,3)
Convolutional: (P, Q)	(3,3), (6,4), (8,4), (8,5)
Helical: (C, h, L)	(3,2,1), (5,4,3), (7,4,3), (7,5,3)
Random: (L)	(100), (200), (300), (400)

The DRN architecture, as depicted in Table 3, features a different structure. It starts with an initial convolutional layer with 64 filters and uses max-pooling to downsample the data. The unique aspect of the DRN is the utilization of residual blocks with dilated convolutions. These residual blocks contain two convolutional layers each, followed by batch normalization and ReLU activation function. Skip connections are added to each block, allowing the network

to capture features at multiple dilation rates. After a series of such blocks, we employ global average pooling to reduce dimensionality, followed by FC layers with ReLU activation and dropout. The output layer, again with softmax activation, consists of 12 units.

1) DATASET GENERATION

In this paper, we classify data that has been interleaved using four distinct interleaver types: convolutional, block, helical interleavers, and random interleavers, each characterized by four different parameters as shown in Table 4. For the interleaver, the input data consists of FEC-encoded data. In this paper, we employed six types of FEC encoders, specifically BCH, RS, block, convolutional, polar, and LDPC. The dataset generation process encompasses several stages, starting with message generation, followed by encoding, interleaving, modulation, the introduction of noise using AWGN model, and concluding with demodulation, as visually represented in Fig. 7. We have augmented the dataset by simulating different signal-to-noise ratio (SNR) levels (in dB): [-10, -5, 0, 5, 10, 15, 20]. Each message signal is passed through an AWGN channel with these SNR values, effectively generating multiple versions of the same message with varying noise levels. To create variations in the input data, we have employed four distinct interleaving techniques. The data augmentation process involves several steps. We generate random messages using Galois Field (GF) function with specified parameters. These random messages are then encoded using different channel coding techniques. Depending on the class index, different interleaving techniques are applied to the encoded data. The interleaved data is then converted into a bipolar format (1 and -1) and transmitted through an AWGN channel with varying SNR levels. The received data is quantized back to binary values (0 and 1). Each sample is labelled according to the interleaving method used. Finally, the processed data, along with the labels, is saved into datasets of various sizes, ultimately forming the complete datasets. The dataset size depends on the parameters, which are given as follows: The total number of samples per class (N), including all SNR levels, is calculated as $N = L \times (M + 1)$, where L denotes the sample size and M denotes the length of SNR array. Thus, the total samples T for all classes are given by $T = C \times N$, where C indicates the number of classes. The final dataset size S will be $S = (T \times i) + 1$, accounting for the last column used for labelling, which represents the class of interleavers, where i indicates the size of the input. We have created datasets for five different input lengths (i): 512, 1024, 2048, 4096, and 16384, with parameters provided as shown in the Table. 4.

Block codes are typically represented as (n, k) , where $r = n - k$ indicates the length of the parity or redundancy bits. The convolutional codes detailed in this paper are designated as (n, k, N_0) , with N_0 signifying the length of the coding constraint. The polar codes outlined in this paper adhere to the format (N, K) , where N denotes the codeword length, consistently taking the form of an exponent of 2, i.e., 2^n

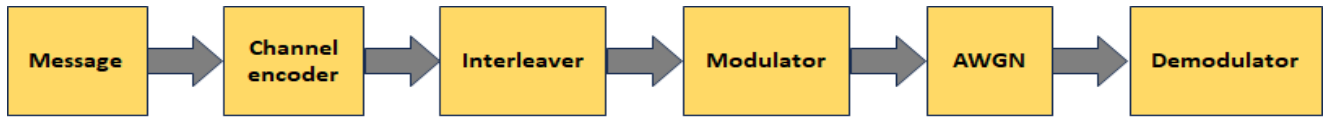


FIGURE 7. Dataset generation process based on simulations illustrating various steps involved in interleaver recognition.

($n \geq 2$), and K signifies the number of message bits. The disparity $N - K$ corresponds to the frozen bits' number with the reliability sequence Q as described in [42]. An LDPC code can be defined in terms of a parity check matrix H with its properties contingent on an expansion factor Z , and this parity matrix can be generated based on the base matrices B [43]. RS codes are non-binary codes that operate over a finite field $GF(2^m)$, where m denotes the number of bits per symbol. RS codes are extensively used in digital media (CDs, DVDs, Blu-rays) and various communication systems (digital television, QR codes). Additionally, they provide strong data integrity for high-reliability applications, such as deep-space communication.

We employ the Hamming code as our block encoder, characterized by a codeword length of $n = 7$, a code dimension of $k = 4$, and a generator matrix $G = [1010011; 1001001; 0011011; 1000101]$. Additionally, we incorporate a BCH code with parameters $n = 31$ and $k = 21$. Our chosen convolutional encoder operates at a rate of $1/2$, featuring a constraint length of $N_0 = 4$ and a generator polynomial $g = [15, 17]$. The fourth encoder in use is a polar encoder based on 5G standard, employing a reliability sequence of $Q = 1024$ for a codeword length of $N = 16$ and a message bit length of $K = 2^n$, where n belongs to the integer values. The fifth encoder is LDPC for which we have utilized base matrices (B) from 5G standard of size 46×68 with an expansion factor $Z = 400$. Finally, we have considered RS encoder as the sixth encoder with code parameters as follows: $n = 7$, $k = 3$, and number of bits per symbol $m = 3$.

TABLE 5. Run time for the CNN model.

Input length	Training time/step	Testing time/step
512	8ms	2ms
1024	10.5ms	3ms
2048	16.5ms	5ms
4096	32ms	9ms
8192	66ms	17ms
16384	144ms	35ms

From Table 4, it is observed that the dataset encompasses a total of 16 distinct interleavers, each with four different sizes, resulting in the generation of 500 sets for each size. In order to replicate real-world scenarios, we introduce noise through varying SNR spanning from -10 dB to 20 dB with 5 dB intervals, alongside an ideal scenario without adding noise. Considering that we generate 8000 samples per size, the overall dataset comprises a total of 1,28,000 samples.

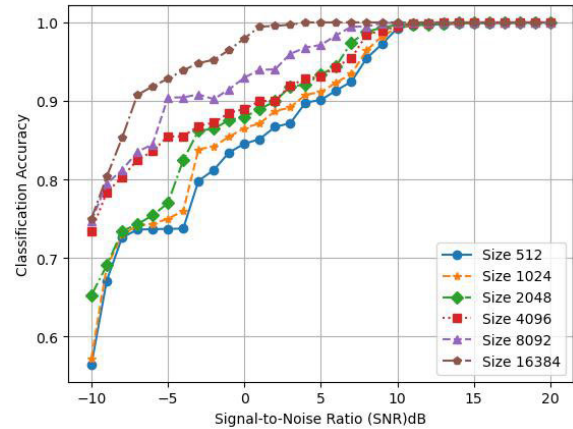


FIGURE 8. Classification accuracy of interleavers from convolutional encoded data across various SNR levels for CNN.

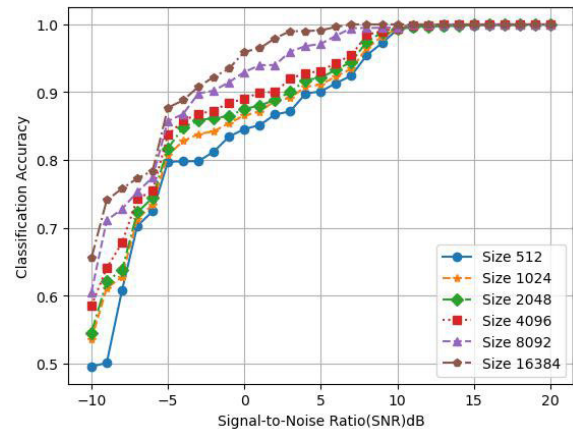


FIGURE 9. Classification accuracy of interleavers from block encoded data across various SNR levels for CNN.

2) INPUT DATA

Keras [44] is a widely used and powerful deep learning framework for Python. In this work, our proposed model is constructed and trained using Keras. We opt for the cross-entropy loss function due to its ability to facilitate faster learning through gradient descent, surpassing the quadratic loss. Our choice of optimization algorithm is Adam [45], with an initial learning rate set to 0.001 ($L_r = 0.001$). As the model's performance reaches a plateau, we gradually reduce the learning rate by a factor of 5. The training and testing of our model were conducted within an environment created in Anaconda Navigator. These operations were

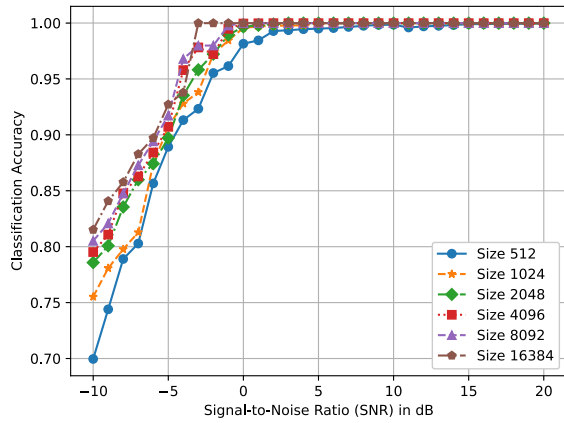


FIGURE 10. Classification accuracy of interleavers from BCH encoded data across various SNR levels for CNN.

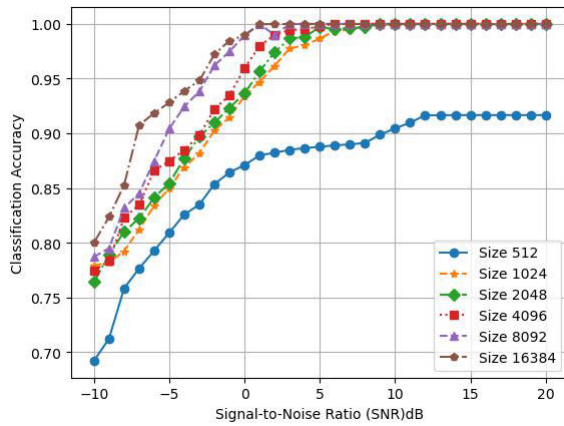


FIGURE 11. Classification accuracy of interleavers from LDPC encoded data across various SNR levels for CNN.

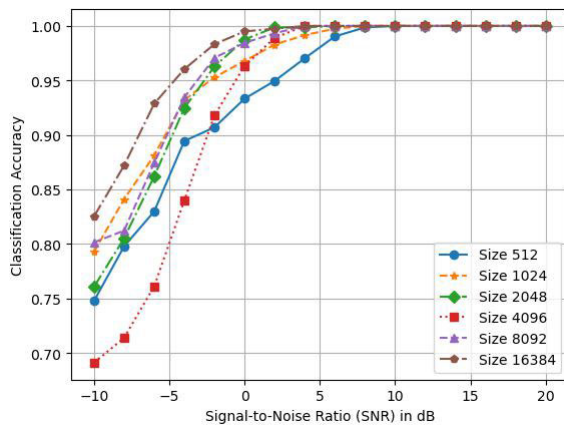


FIGURE 12. Classification accuracy of interleavers from polar encoded data across various SNR levels for CNN.

carried out on a PC equipped with a 64GB RAM and a core i9 processor. The experimental setup involves the generation of 10,000 samples, each having a length of 1000 and stored in int32 format. In the datasets, 80%

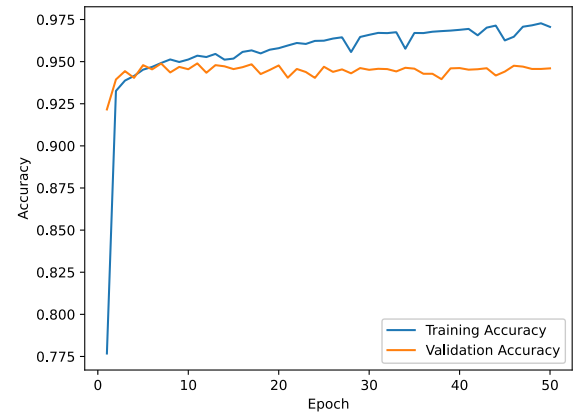


FIGURE 13. Training versus validation curve for CNN model.

of the samples were allocated for training the interleaver recognition model, while the remaining 20% were reserved for evaluating the performance. Our primary objective is to determine whether the CNN and DRN models outperform other algorithms in terms of recognition or classification accuracy. We assess the accuracy, precision, recall, and F1 score by calculating the percentage of correctly predicted outcomes. Specifically, for a given category of interleaver, a true positive (TP) signifies that our model accurately assigns the interleaver to its respective category. Conversely, a false positive (FP) denotes that the interleaver is misclassified. Consequently, the accuracy of each category is defined as

$$\text{Accuracy(in \%)} = \frac{TP}{TP + FP} \times 100 \quad (4)$$

whereas precision is the ratio of correctly predicted positive observations to the total predicted positives, recall is the ratio of correctly predicted positive observations to all observations in the actual class, and the F1 score is the weighted average of precision and recall.

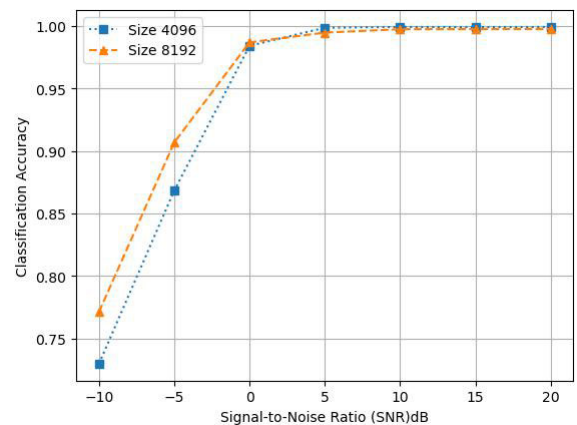


FIGURE 14. Classification accuracy of interleavers from convolutional encoded data across various SNR levels for DRN.

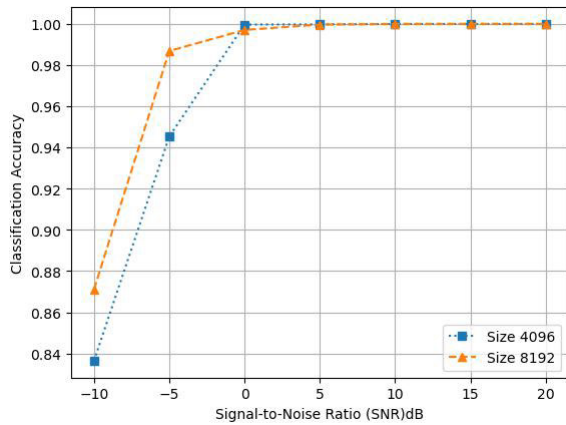


FIGURE 15. Classification accuracy of interleavers from polar encoded data across various SNR levels for DRN.

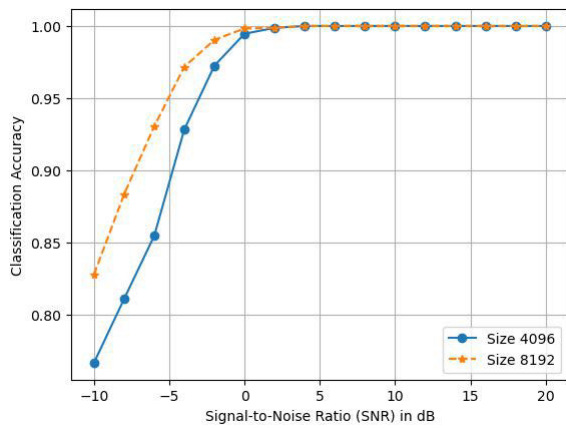


FIGURE 16. Classification accuracy of interleavers from block encoded data across various SNR levels for DRN.

IV. RESULTS AND DISCUSSIONS

A. ANALYZING THE IMPACT OF INPUT LENGTH

The length of the input sample plays a pivotal role in the blind recognition of interleavers. As the input length increases, both computational complexity and training time exhibit simultaneous growth. To investigate the influence of input sample length on the model's performance, we have conducted training of the CNN model using samples of varying lengths such as 512, 1024, 2048, 4096, 8192, and 16384. Further, in DRN model, we have used samples of varying lengths, which include 4096 and 8192, while considering a wide range of SNRs from -10 dB to 20 dB.

The experimental outcomes highlight the impact of varying input lengths within the test dataset. Specifically, the model demonstrates improved average recognition accuracy, as the input sample length increases, with a noticeable enhancement observed with each doubling of the input sample length.

Table 5 presents the average run time for training and testing per example across different input lengths for CNN model. Notably, the results reveal that increasing the sample length by a factor of two corresponds to a relatively modest

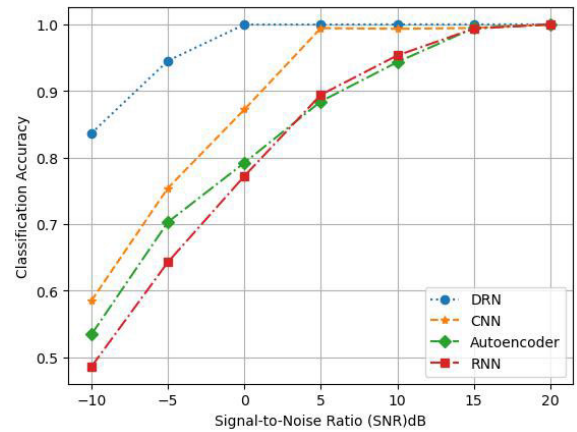


FIGURE 17. Comparison of classification accuracy from block encoded data across various SNR levels among autoencoder, CNN, RNN, and DRN.

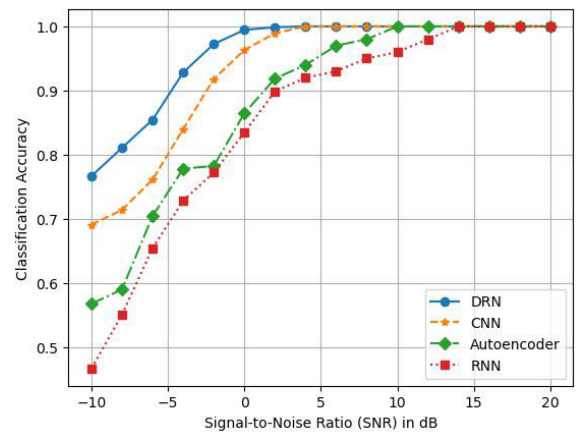


FIGURE 18. Comparison of classification accuracy from polar encoded data across various SNR levels among autoencoder, CNN, RNN, and DRN.

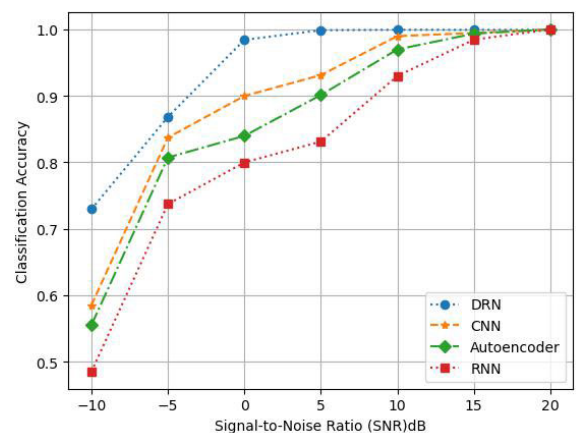


FIGURE 19. Comparison of classification accuracy from convolutional encoded data across various SNR levels among autoencoder, CNN, RNN, and DRN.

increase of less than 2.5 times in training and testing time for each sample. Specifically, when the input samples are

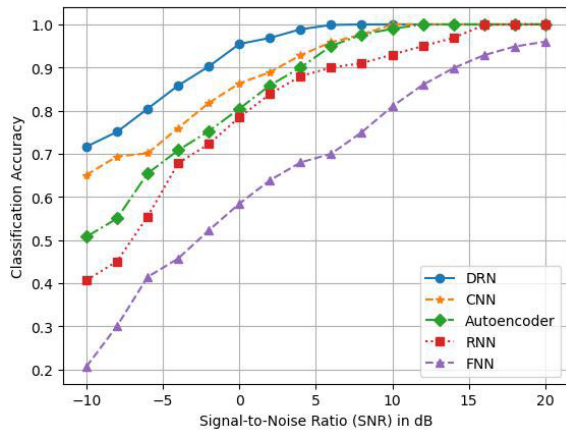
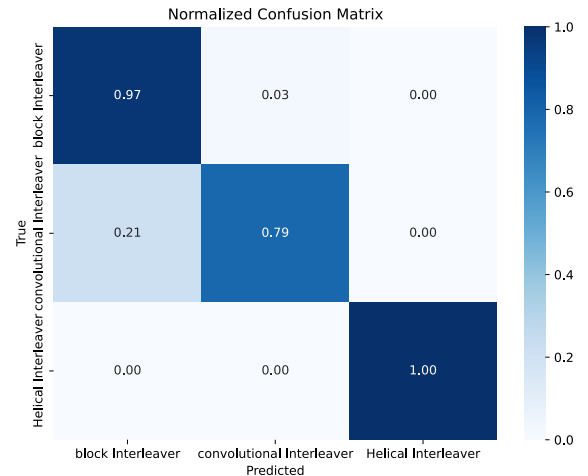


FIGURE 20. Comparison of classification accuracy from RS encoded data across various SNR levels among autoencoder, CNN, RNN, DRN, and FNN.

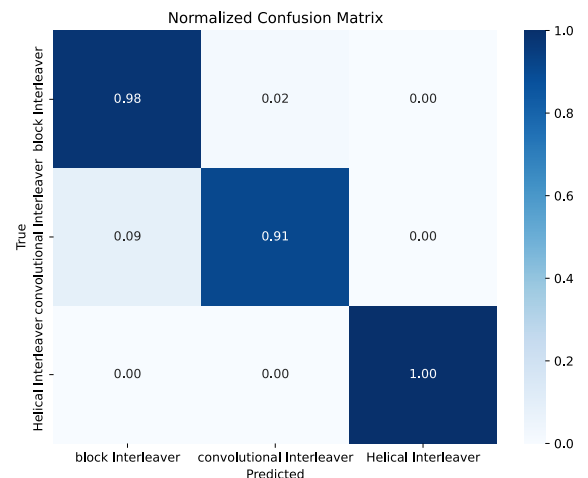
of length 16384, the training time per sample is measured at 144 ms, a duration deemed acceptable for real-world applications. This phenomenon is attributed to the fact that changes in the input sample lengths do not significantly impact the network architecture or the number of model parameters, resulting in minimal effects on the computational complexity. Consequently, we have chosen an input sample length of 16,384, which yields the highest recognition accuracy. Moreover, in scenarios where the application environment remains constant, the model requires training only once, streamlining the operational process.

B. ACCURACY OF INTERLEAVER CLASSIFICATION VIA CNN MODEL

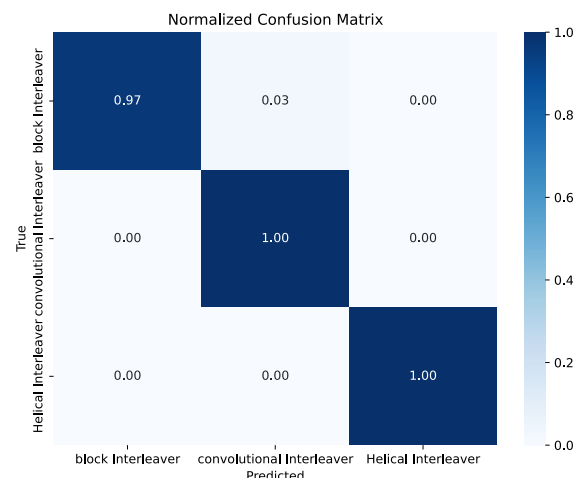
Figs. 8, 9, 10, 11, and 12 present the classification results for block, convolutional, and helical interleavers across various parameter settings for the channel encoder. From Fig. 8, where the convolutional encoder serves as the input to the interleaver, we observe that the classification accuracy of the interleaver consistently exceeds 95% across a range of SNR values beyond 5 dB for all input lengths. Notably, as SNR levels increase, the accuracy also exhibits a noticeable upward trend, as clearly depicted in the figures. Similarly in Fig. 9, we observe that the classification accuracy of the interleaver also surpasses 95% beyond 5 dB SNR level for all input lengths except the input length of 512. In this scenario, we have utilized a block encoder as an input to the interleaver. Similarly, in Fig. 10, where the encoder utilized is BCH, the accuracy surpasses 95% beyond 0 dB. In Figs. 11 and 12, it is evident that the accuracy consistently exceeds 95% for all input lengths after 5 dB SNR level, except for an input length of 512, while using LDPC and polar encoders, respectively. This leads to the conclusion that increasing the input length results in higher accuracy within a specific SNR range. In the context of the obtained results, it is inferred that BCH encoding yields higher classification accuracy of interleavers compared to alternative



(a) SNR -4 dB



(b) SNR 0 dB



(c) SNR 4 dB

FIGURE 21. Confusion matrices illustrating the classification accuracy of interleavers under convolutional encoding at different SNR levels for CNN.

methods, specifically under 5 dB SNR levels. This is because, the compatibility of BCH-encoded data with the inherent

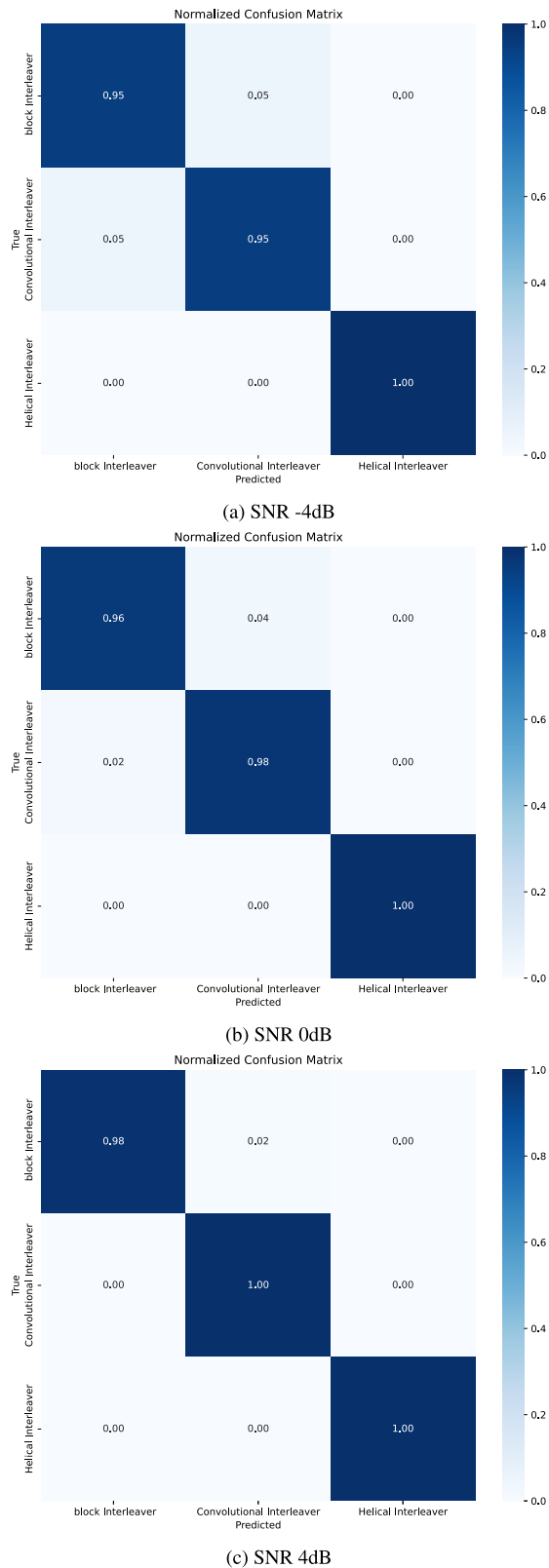


FIGURE 22. Confusion matrices illustrating the classification accuracy of interleavers under convolutional encoding at different SNR levels for DRN.

strengths of CNN architectures, such as pattern recognition, translation invariance, local connectivity, and adaptability

to noisy conditions, collectively contribute to the observed higher classification accuracy. The synergy between BCH encoding and CNN architecture results in a model that effectively exploits the unique features introduced by BCH codes for improved performance in interleaver classification task.

In Fig. 13, the training and validation curves for LDPC-encoded interleavers exhibit a consistent upward trend, indicating effective model learning across epoch. Noteworthy is the model's ability to generalize effectively, which is evident in the close alignment of the validation curve with the training curve throughout the training process [46].

C. ACCURACY OF INTERLEAVER CLASSIFICATION VIA DRN MODEL

Figs. 14, 15, and 16 present the classification accuracy results for interleaver classification across three channel encoders, i.e. block, convolutional, and polar, respectively, utilizing DRN architecture. The observed trend indicates that the accuracy reaches 100% after 5 dB for two specific input sample lengths: 4096 and 8192. We selected only two input sample lengths for our experiments with the DRN, as the training and testing processes are time-consuming. By focusing on these two input lengths, we aim to gain insights into the working of the DRN and its classification accuracy. The choice of these specific input lengths allows us to efficiently assess the network's performance without the extended time requirements associated with training and testing across a broader range of input lengths. Notably, for polar and block encoding (i.e. Fig. 15 and Fig. 16), higher accuracy is observed at 0 dB SNR compared to the convolutional encoder (i.e. Fig. 14). Furthermore, the superior performance of polar and block encoding in classifying the interleavers effectively at 0 dB SNR underscores their resilience in challenging communication conditions, outperforming the convolutional encoder in these specific scenarios.

D. COMPARATIVE ANALYSIS WITH RNN AND AUTOENCODER

In this sub-section, we have compared the classification accuracy of the proposed CNN and DRN models with other deep learning models such as autoencoder [47], FNN [48], and RNN [49] available in the literature. In our comparative analysis of classifying interleavers using four deep learning models, i.e. CNN, DRN, autoencoder, and RNN, we observe that DRN achieves the highest accuracy, followed by CNN, autoencoder, and RNN, while maintaining a similar level of complexity, as depicted in Figs. 17, 18 and 19. We assume an input sample length of 4096 for all three models and plotted the curves at SNR levels ranging from -10 to 20 dB. For comparative analysis, we employ different encoders for each figure such as block encoder in Fig. 17, polar encoder in Fig. 18, and convolutional encoder in Fig. 19. However, it is worth noting that DRN incurs longer training and testing times, which is almost 30 times more compared to

TABLE 6. Performance comparison of various deep learning models.

Model	Training time/step	Total time	training	Testing time/step	Total time	testing	F1 Score	Recall	Precision
DRN	1s	500min		346ms	173min		0.9764	0.9867	0.9802
CNN	32ms	16min		9ms	4.5min		0.9596	0.9607	0.9708
Autoencoder	30ms	15min		8ms	4min		0.9215	0.9230	0.9223
RNN	14ms	7min		3ms	0.5min		0.9013	0.9123	0.9100

CNN, autoencoder, and RNN models as shown in Table 6. Consequently, we conclude that CNN holds an advantage over DRN, autoencoder, and RNN for interleaver recognition task with slightly lower accuracy than DRN, but significantly reduced training and testing time as seen in Table 6.

We have performed the classification task using the RS encoded data with four types of interleavers, including random interleaver, at an input sample length of 4096, comparing the results across all five models, i.e. DRN, CNN, autoencoder, RNN, and FNN [48], as shown in Fig. 20. In this scenario as well, the DRN model achieves better accuracy, though it requires longer training and testing times. This indicates the generalization capability of the proposed model, which can classify numerous types of interleavers and achieves an accuracy greater than 95% at a certain SNR level.

Figs. 21a, 21b, and 21c depict the confusion matrices corresponding to the SNR levels of -4 dB, 0 dB, and 4 dB, respectively, for the classification of interleavers using a convolutional encoder with an input length of 4096 assuming CNN model. In these matrices, the horizontal axis represents the true values, while the vertical axis signifies the predicted values. Upon scrutinizing the confusion matrices, it is observed that as the SNR level increases from -4 dB to 4 dB, the classification accuracy experiences a significant improvement. Notably, the helical interleaver exhibits nearly 100% efficiency across all three SNR levels, outperforming other interleavers. This superior performance can be attributed to the distinctive patterns introduced by helical interleavers. The specific arrangement of bits in a spiral pattern creates discernible features that facilitate accurate classification by the CNN model.

Figs. 22a, 22b, and 22c portray the confusion matrices generated by our DRN model at the SNR levels of -4 dB, 0 dB, and 4 dB, respectively. The dark blue line within these matrices signifies the classification accuracy of all interleavers. By analyzing these figures, it is inferred that the DRN model consistently outperforms the CNN model in terms of accuracy at each SNR level. Notably, significant difference in the classification of convolutional interleavers is observed, highlighting the efficacy of the DRN model compared to CNN model in this context, but with higher time complexity as discussed before.

We have also plotted a confusion matrix at input sample length of 4096 and 5 dB SNR level. Fig. 23 depicts the confusion matrix of the DRN model for RS encoded interleavers. It clearly shows that all four interleavers are

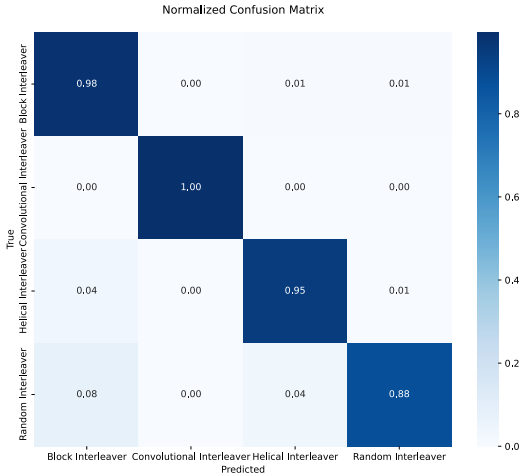


FIGURE 23. Confusion matrices illustrating the classification accuracy of interleavers under RS encoding at 5 dB SNR level for DRN.

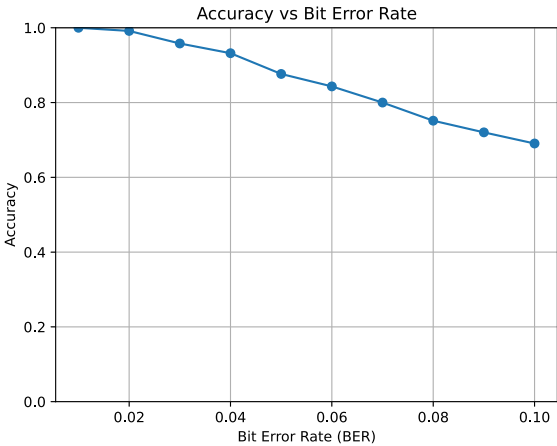


FIGURE 24. Classification accuracy versus BER with input size 16384.

classified with good accuracy, almost above 90%, except for the random interleaver, which is often confused with the helical interleaver. This confusion arises due to the helical interweaver’s complex pattern, which disperses adjacent data points in an apparently random fashion. This intricate arrangement makes the output of a helical interleaver resemble that of a random interleaver, unlike the more systematic patterns produced by block and convolutional interleavers.

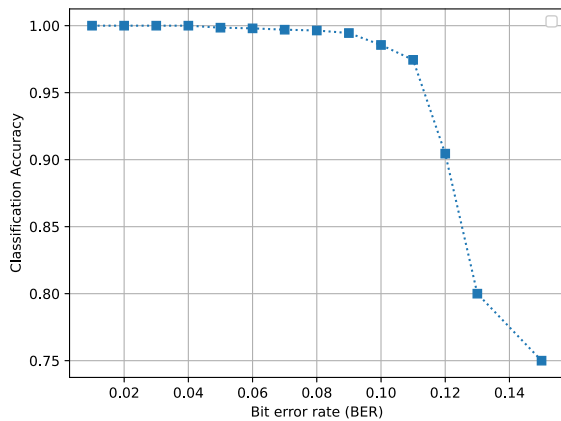


FIGURE 25. Classification accuracy of three different convolutional interleavers from BCH encoded data across various BER levels for CNN.

TABLE 7. Simulation parameters.

FEC encoder	Convolutional interleaver parameters
BCH (7,4)	(3,2)
BCH (7,4)	(4,2)
BCH (7,4)	(6,2)

Here, we compare our proposed 1-D CNN model with the existing rank-based algorithm introduced in [19] and [26] using GJETP technique. The rank-based algorithm identifies interleaver parameters through the rank deficiency method, determining block interleaver parameters from block encoded data across various bit error rates (BERs), as depicted in Fig. 15 of [26]. It achieves 80% accuracy at $\text{BER} = 2 \times 10^{-5}$. We classify block interleavers from block encoded data for different BER values using our proposed CNN model, as illustrated in Fig. 24, achieving improved accuracy (around 90%) at the same BER value. Furthermore, we also estimate the parameters of BCH encoded convolutional interleavers using our CNN model, as detailed in Table 7, and compare its accuracy as shown in Fig. 25 with the accuracy of the existing GJETP model proposed in [19] (Fig. 12). Our findings demonstrate that our model outperforms the existing one at the same BER, specifically at $\text{BER} = 0.1$. In our model, which employs BCH encoded data and various convolutional interleavers, the BCH encoding techniques are effective up to a specific BER value, which is 0.1 in this case. Beyond this BER value, the noise overwhelms the CNN's pattern recognition capacity, resulting in a sharp decline in the classification accuracy.

V. CONCLUSION

In this paper, we proposed deep-learning-based CNN and DRN models to recognize the correct interleaver among block, convolutional, random, and helical interleavers. The interleaver recognition task plays a vital role in non-cooperative military communication systems, spectral-efficient adaptive-modulation-and-coding-based systems,

and design of intelligent receiver for 5G and beyond wireless systems. For performing the interleaver classification, we assumed that the datasets were encoded using block, convolutional, BCH, LDPC, RS, and polar encoders. Both CNN and DRN models consistently achieved classification accuracy exceeding 95% across different input lengths, showcasing their robustness. Notably, higher classification accuracy was achieved with an increase in input lengths. Apart from the accuracy metric, the models were also evaluated using metrics such as the F1 score, recall, and precision values which provided a more comprehensive understanding of the model's performance. Furthermore, from our comparative analysis involving autoencoder, CNN, DRN, FNN, and RNN deep learning models, it was determined that the DRN model performed the best in terms of classification accuracy, F1 score, precision, and recall, although with higher time complexity. Moreover, the CNN model performed better than the autoencoder, FNN, and RNN models in terms of accuracy, precision, recall, and F1 score, and also exhibited efficiency with reduced training and testing times compared to DRN. From the confusion matrices, it was inferred that the helical interleaver exhibits better accuracy, outperforming other interleavers. In addition, it was also noticed that the DRN model aids in classifying convolutional interleavers with higher accuracy compared to the CNN model. Finally, we compared our CNN model with the existing GJETP method in terms of classification accuracy and observed that our model outperforms the existing one. Our future work is focused towards joint classification of the channel encoders, modulation schemes, and interleavers using various deep learning models as well as experimental validation of the proposed deep learning models using real datasets generated from software-defined radio (SDR) units.

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